

Session 1.3 — The Transformer Architecture

RNN, Transformer Architecture and Attention

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Session Overview

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This session completes our study of sequence modeling with RNNs and LSTMs, then transitions to attention-based models and the Transformer architecture.

We begin with Recurrent Neural Networks (RNNs), examining how hidden states model dependencies across time steps and how training works via Backpropagation Through Time (BPTT). We analyze why vanilla RNNs suffer from vanishing and exploding gradients, limiting long-range dependencies, and how LSTM networks address this through gated mechanisms (input, forget, and output gates) and a cell state.

Building on our earlier discussion of word representations, we move from static embeddings (Word2Vec) to contextual embeddings (ELMo), where a word's meaning depends on its context. This motivates the central question: how can models capture relationships between all tokens in a sequence, efficiently and in parallel?

This leads to the attention mechanism, which lets each token directly access information from every other token. We then take a deep dive into the Transformer architecture — its encoder-decoder structure, self-attention using Query, Key, and Value representations, scaled dot-product

attention, multi-head attention, positional encoding, feed-forward networks, and residual connections with layer normalization.

Key components covered:

- RNN hidden states, BPTT, and gradient problems
- LSTM gates and cell state
- Static (Word2Vec) vs contextual (ELMo) embeddings
- Self-Attention (Q, K, V) and Scaled Dot-Product Attention
- Multi-Head Attention and Positional Encoding
- FFN, Residual Connections & Layer Normalization

Learning Outcomes

By the end of this lecture, participants should be able to:

- Explain how RNNs model sequences using hidden states and are trained via BPTT.
- Analyze the vanishing/exploding gradient problems and how LSTM gating mechanisms address them.
- Distinguish static embeddings (Word2Vec) from contextual embeddings (ELMo).
- Explain the intuition behind attention and self-attention, including the role of Q, K, and V.
- Describe the Transformer architecture and how scaled dot-product attention works.
- Explain why multi-head attention and positional encoding are needed.
- Identify the role of FFNs, residual connections, and layer normalization in stabilizing training.

Prerequisites

- Core ML concepts: training, loss functions, and model evaluation.
- Neural network fundamentals and basic linear algebra/calculus.
- CNNs and word representations (One-Hot, Word2Vec) from the previous session.
- Basic Python programming and analytical thinking.

Session Takeaway

By the end of this session, participants will develop a rigorous and structured understanding of attention-based models and the Transformer architecture as the foundation of modern sequence modeling. They will gain:

- A clear understanding of attention and self-attention, including how tokens interact using Query, Key, and Value representations.

- Insight into the design of the Transformer architecture, and how it enables parallel processing over sequences.
- A deep conceptual understanding of multi-head attention and positional encoding, and their role in capturing relationships and preserving order.
- An appreciation of how Transformers handle long-range dependencies efficiently, overcoming limitations of earlier sequential approaches.